

4.24

Suppose that the bandlimited differentiator of Example 4.4 has input $x_c(t) = \cos(\Omega_0 t)$ with $\Omega_0 < \pi/T$. In this problem, we wish to verify that the continuous-time signal reconstructed from the output of the bandlimited differentiator is indeed the derivative of $x_c(t)$.

- The sampled input will be $x[n] = \cos(\omega_0 n)$, where $\omega_0 = \Omega_0 T < \pi$. Determine an expression for $X(e^{j\omega})$ that is valid for $|\omega| \leq \pi$.
- Now use Eq. (4.46) to determine the DTFT of $Y(e^{j\omega})$, the output of the discrete-time system.
- From Eq. (4.32) determine $Y_r(j\Omega)$, the continuous-time Fourier transform of the output of the D/C converter.
- Use the result of (c) to show that

$$y_r(t) = -\Omega_0 \sin(\Omega_0 t) = \frac{d}{dt} [x_c(t)].$$

Remarks:

$$(1) Y_r(j\Omega) = H_r(j\Omega)Y(e^{j\Omega T})|_{\omega=\Omega T} = H_r(j\Omega)Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \frac{\pi}{T} \\ 0, & \text{otherwise} \end{cases} \quad (4.32)$$

$$(2) H(e^{j\omega}) = j\frac{\omega}{T}, \quad |\omega| < \pi. \quad (4.46)$$

Sol (a) $\cos(\omega_0 n) = \frac{1}{2}(e^{j\omega_0 n} + e^{-j\omega_0 n})$, $e^{j\omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} X(e^{j(\omega-\omega_0)})$

$\xleftrightarrow{\text{DTFT}} 2\pi\delta(\omega) \therefore X(e^{j\omega}) = \frac{1}{2}[2\pi\delta(\omega-\omega_0) + 2\pi\delta(\omega+\omega_0)] = \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$

(b) $Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}) = \frac{j\omega}{T}[\pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)] = \frac{j\omega_0\pi}{T}\delta(\omega-\omega_0) - \frac{j\omega_0\pi}{T}\delta(\omega+\omega_0)$

(c) $Y_r(j\Omega) = H_r(j\Omega)Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \frac{\pi}{T} \\ 0, & \text{otherwise} \end{cases}$
 $= j\omega_0\pi\delta(\Omega T - \omega_0) - j\omega_0\pi\delta(\Omega T + \omega_0) = j\omega_0\pi\delta[(\Omega - \frac{\omega_0}{T})T] - j\omega_0\pi\delta[(\Omega + \frac{\omega_0}{T})T]$

(d) $\delta(\Omega T) \xleftrightarrow{\text{Inverse F.T.}} \frac{1}{2\pi T}$, $y_r(t) = \frac{j\omega_0}{2T}e^{j\frac{\omega_0 t}{T}} - \frac{j\omega_0}{2T}e^{-j\frac{\omega_0 t}{T}}$
 $= -\Omega_0 \left(\frac{e^{j\frac{\omega_0 t}{T}} - e^{-j\frac{\omega_0 t}{T}}}{j2} \right) = -\Omega_0 \sin(\Omega_0 t)$

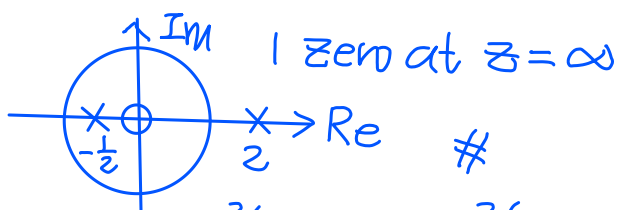
5.8

A causal LTI system is described by the difference equation

$$y[n] = \frac{3}{2}y[n-1] + y[n-2] + x[n-1].$$

- Determine the system function $H(z) = Y(z)/X(z)$ for this system. Plot the poles and zeros of $H(z)$, and indicate the ROC.
- Determine the impulse response of the system.
- You should have found the system to be unstable. Determine a stable (noncausal) impulse response that satisfies the difference equation.

Sol (a) $Y(z) = \frac{3}{2}z^{-1}Y(z) + z^{-2}Y(z) + z^{-1}X(z)$
 $\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{1 - \frac{3}{2}z^{-1} - z^{-2}} = \frac{z^{-1}}{(1-2z^{-1})(1+\frac{1}{2}z^{-1})}, |z| > 2$



(b) $H(z) = \frac{z^5}{1-2z^{-1}} + \frac{-z^5}{1+\frac{1}{2}z^{-1}} \therefore h[n] = \frac{2}{5}[(2)^n - (-\frac{1}{2})^n]u[n]$ #

(c) By ROC: $\frac{1}{2} < |z| < 2$, $h[n] = -\frac{2}{5}(2)^n u[n-1] - \frac{2}{5}(-\frac{1}{2})^n u[n]$ #

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A discrete-time causal LTI system has the system function

$$H(z) = \frac{(1 + 0.2z^{-1})(1 - 9z^{-2})}{(1 + 0.81z^{-2})}$$

(a) Is the system stable?

(b) Determine expressions for a minimum-phase system $H_1(z)$ and an all-pass system $H_{ap}(z)$ such that

$$H(z) = H_1(z)H_{ap}(z).$$

(sd) (a) stable, pole: $z = \pm j0.9$ are inside the unit circle.

(b) $H(z) = \frac{1+0.2z^{-1}}{1+0.81z^{-2}} \cdot \frac{1-9z^{-2}}{1} = \frac{(1+0.2z^{-1})(1-\frac{1}{9}z^{-2})}{1+0.81z^{-2}} \frac{1-9z^{-2}}{1-\frac{1}{9}z^{-2}}$

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$= H_1(z)H_{ap}(z)$ #

Consider the class of discrete-time filters whose frequency response has the form

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{-j\alpha\omega},$$

where $|H(e^{j\omega})|$ is a real and nonnegative function of ω and α is a real constant. As discussed in Section 5.7.1, this class of filters is referred to as linear-phase filters.

Consider also the class of discrete-time filters whose frequency response has the form

$$H(e^{j\omega}) = A(e^{j\omega})e^{-j\alpha\omega + j\beta},$$

where $A(e^{j\omega})$ is a real function of ω , α is a real constant, and β is a real constant. As discussed in Section 5.7.2, filters in this class are referred to as generalized linear-phase filters.

For each of the filters in Figure P5.15, determine whether it is a generalized linear-phase filter. If it is, then find $A(e^{j\omega})$, α , and β . In addition, for each filter you determine to be a generalized linear-phase filter, indicate whether it also meets the more stringent criterion for being a linear-phase filter.

generalized:

$A(\omega)$ 可能 > 0

linear:

$A(\omega)$ 一定 > 0

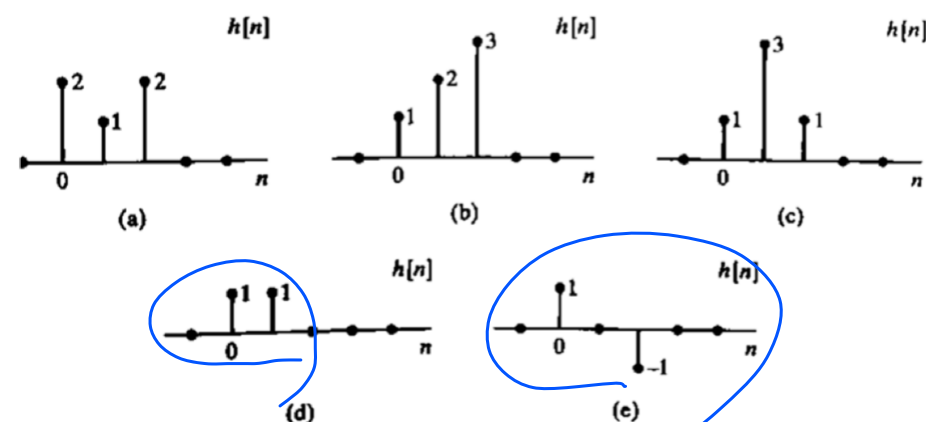


Figure P5.15

$$\text{sol} \rightarrow (a) H(e^{j\omega}) = 2 + e^{-j\omega} + 2e^{-j2\omega} = e^{-j\omega}(2e^{j\omega} + 1 + 2e^{-j2\omega}) = e^{-j\omega}(1 + 4\cos\omega)$$

$A(\omega) = 1 + 4\cos\omega$, $\alpha=1$, $\beta=0 \Rightarrow$ Generalized but not linear

(b) no even/odd symmetry, $\therefore$ not generalized linear-phase filters

$$(c) H(e^{j\omega}) = 1 + 3e^{-j\omega} + e^{-j2\omega} = e^{-j\omega}(e^{j\omega} + 3 + e^{-j2\omega}) = e^{-j\omega}(3 + 2\cos\omega)$$

$A(\omega) = 3 + 2\cos\omega$, $\alpha=1$, $\beta=0 \Rightarrow$ Generalized & linear

$$(d) \text{ even symmetry, } H(e^{j\omega}) = 1 + e^{-j2\omega} = e^{-j\frac{\omega}{2}}(e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}}) = e^{-j\frac{\omega}{2}} \cdot 2\cos\frac{\omega}{2}$$

$A(\omega) = 2\cos\frac{\omega}{2}$, $\alpha=\frac{1}{2}$, $\beta=0 \Rightarrow$ Generalized but not linear

$$(e) \text{ odd symmetry, } H(e^{j\omega}) = 1 - e^{-j2\omega} = e^{-j\omega}(e^{j\omega} - e^{-j\omega}) = e^{-j\omega} \cdot 2j\sin\omega$$

$$= 2\sin\omega e^{-j\omega + j\frac{\pi}{2}}, A(\omega) = 2\sin\omega, \alpha=1, \beta=\frac{\pi}{2} \Rightarrow \text{Generalized but not linear.}$$

Matlab:

Plot the frequency response and group delay of

$$H(z) = \frac{0.001836(1+z^{-1})^4}{(1-1.5548z^{-1}+0.6493z^{-2})(1-1.4996z^{-1}+0.8482z^{-2})}$$

```
>> K = input('Number of frequency points=')
Number of frequency points=256
K =
    256
>> subplot(3,1,3);
>> plot(w,grd);
>> xlabel('Frequency \omega')
>> ylabel('Samples')
>> title('Group Delay')
```

```
>> w = 0:pi/(K-1):pi;
>> n1 = [1 1];
>> num = 0.001836 * conv(conv(conv(n1,n1),n1),n1);
>> d1 = [1 -1.5548 0.6493];
>> d2 = [1 -1.4996 0.8482];
>> den = conv(d1,d2);
>> h = freqz(num, den,w);
>> grd = grpdelay(num, den,w);
```

```
>> subplot(3,1,1);
>> plot(w,angle(h));
>> xlabel('Frequency \omega')
>> ylabel('Radian')
>> title('Phase spectrum')
```

```
>> subplot(3,1,2);
>> plot(w,abs(h));
>> xlabel('Frequency \omega')
>> ylabel('Amplitudde')
>> title('Magnitude spectrum')
```

